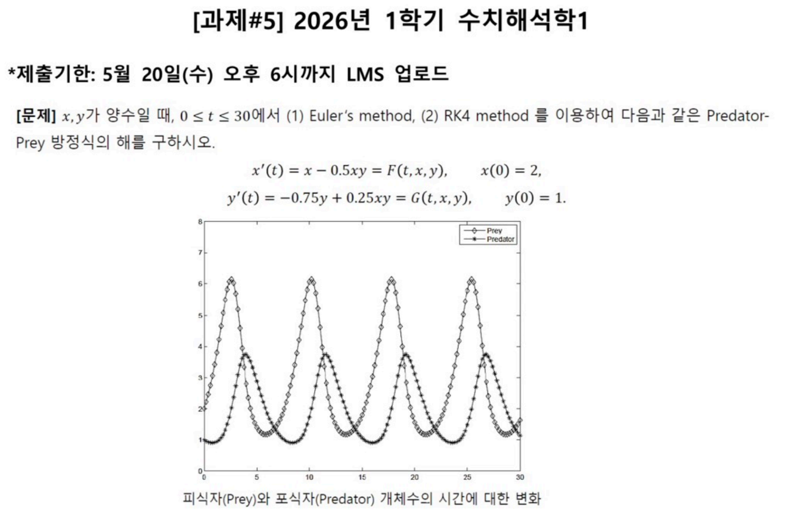


[과제#5] 2026년 1학기 수치해석학1

***제출기한: 5월 20일(수) 오후 6시까지 LMS 업로드**

[문제] x, y가 양수일 때, $0 \le t \le 30$에서 (1) Euler's method, (2) RK4 method 를 이용하여 다음과 같은 Predator-Prey 방정식의 해를 구하시오.

$$x'(t) = x - 0.5xy = F(t, x, y), \qquad x(0) = 2,$$
$$y'(t) = -0.75y + 0.25xy = G(t, x, y), \qquad y(0) = 1.$$

Prey
Predator
8
7
6
5
4
3
2
1
0
0
5
10
15
20
25
30

피식자(Prey)와 포식자(Predator) 개체수의 시간에 대한 변화

학번 : 20221414

이름 : 김민재.

1). 오일러 방식.

$y(t+h) = y(t) + y'(t)h + O(h^2)$

$y(t+h) \approx y(t) + y'(t)h.$

$w(t_i+h) = y(t_i) + y'(t_i)h.$

$x(t+h) = x(t) + x'(t)h + O(h^2)$

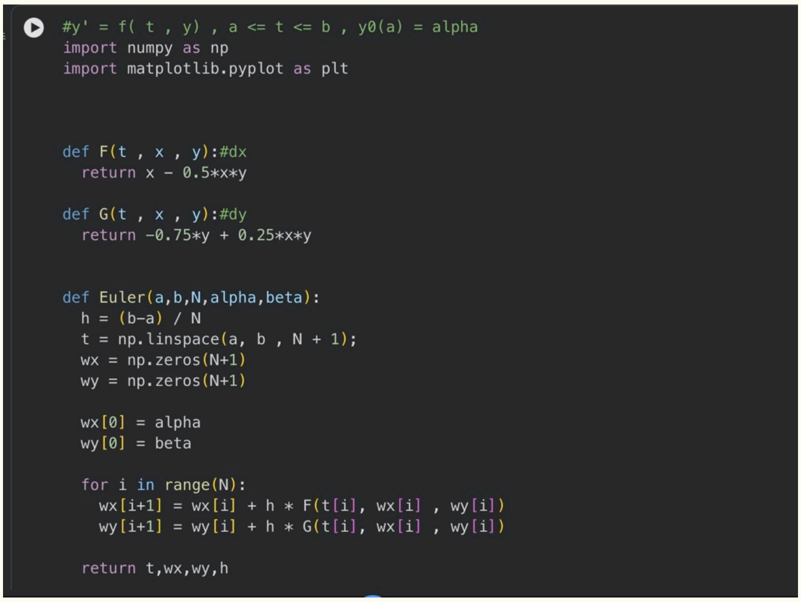
$x(t+h) \approx x(t) + x'(t)h.$

$w_x(t_i+h) = x(t_i) + x'(t_i)h.$

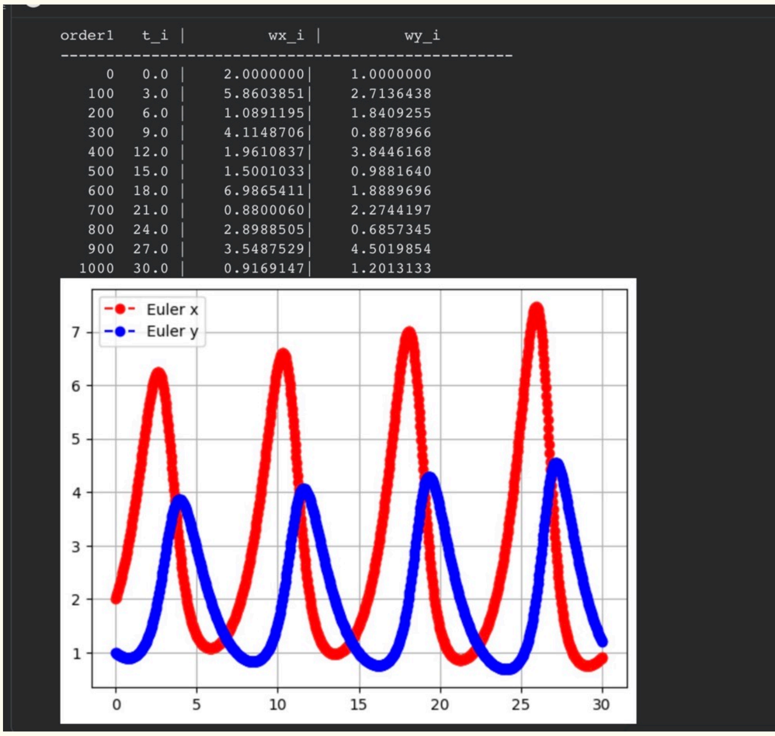
$w_y(t_i+h) = w_y(t_i) + (-0.75\, w_y(t_i) + 0.25\, w_x(t_i)\cdot w_y(t_i))h.$

$w_x(t_i+h) = w_x(t_i) + (w_x(t_i) - 0.5\, w_x(t_i)\cdot w_y(t_i))h.$

$(a, b) = (0, 30), N = 1000.$



```
#y' = f( t , y) , a <= t <= b , y0(a) = alpha
import numpy as np
import matplotlib.pyplot as plt



def F(t , x , y):#dx
  return x - 0.5*x*y

def G(t , x , y):#dy
  return -0.75*y + 0.25*x*y


def Euler(a,b,N,alpha,beta):
  h = (b-a) / N
  t = np.linspace(a, b , N + 1);
  wx = np.zeros(N+1)
  wy = np.zeros(N+1)

  wx[0] = alpha
  wy[0] = beta

  for i in range(N):
    wx[i+1] = wx[i] + h * F(t[i], wx[i] , wy[i])
    wy[i+1] = wy[i] + h * G(t[i], wx[i] , wy[i])

  return t,wx,wy,h
```



```
order1   t_i |        wx_i |          wy_i
--------------------------------------------------
     0   0.0 |    2.0000000|    1.0000000
   100   3.0 |    5.8603851|    2.7136438
   200   6.0 |    1.0891195|    1.8409255
   300   9.0 |    4.1148706|    0.8878966
   400  12.0 |    1.9610837|    3.8446168
   500  15.0 |    1.5001033|    0.9881640
   600  18.0 |    6.9865411|    1.8889696
   700  21.0 |    0.8800060|    2.2744197
   800  24.0 |    2.8988505|    0.6857345
   900  27.0 |    3.5487529|    4.5019854
  1000  30.0 |    0.9169147|    1.2013133
```

(2). RK4 method.

$$F(t,x,y) = x - 0.5xy \rightarrow x'(t) \quad x(0)=2$$

$$G(t,x,y) = -0.75y + 0.25xy \rightarrow y'(t) \quad y(0)=1$$

$$y(t+h) \approx y(t) + \frac{1}{6}(k_{y_1} + 2k_{y_2} + 2k_{y_3} + k_{y_4})$$

$$w_y(t_i+h) = w_y(t_i) + \frac{1}{6}(k_{y_1} + 2k_{y_2} + 2k_{y_3} + k_{y_4})$$

$$x(t+h) \approx x(t) + \frac{1}{6}(k_{x_1} + 2k_{x_2} + 2k_{x_3} + k_{x_4})$$

$$w_x(t_i+h) = w_x(t_i) + \frac{1}{6}(k_{x_1} + 2k_{x_2} + 2k_{x_3} + k_{x_4})$$

$$k_{y_1} = hG(t_i, w_x(t_i), w_y(t_i))$$
$$k_{x_1} = hF(t_i, w_x(t_i), w_y(t_i))$$

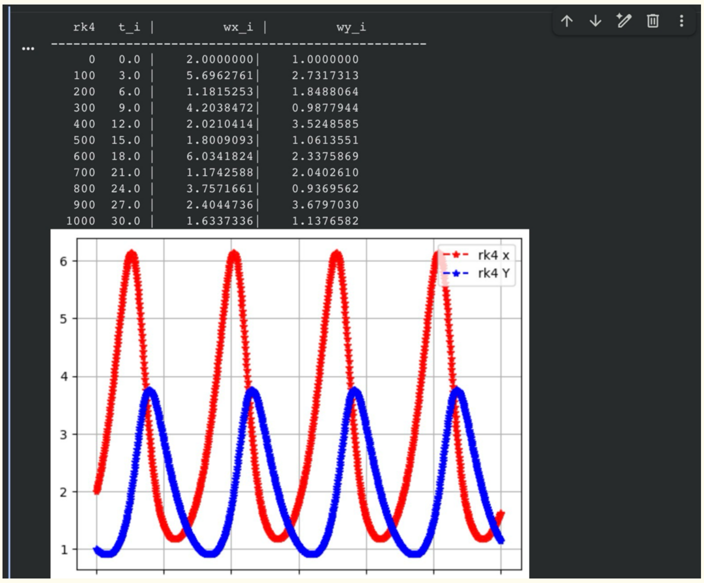
$$k_{y_2} = hG(t_i + \tfrac{h}{2}, w_x(t_i) + \tfrac{1}{2}k_{x_1}, w_y(t_i) + \tfrac{1}{2}k_{y_1})$$
$$k_{x_2} = hF(t_i + \tfrac{h}{2}, w_x(t_i) + \tfrac{1}{2}k_{x_1}, w_y(t_i) + \tfrac{1}{2}k_{y_1})$$

$$k_{y_3} = hG(t_i + \tfrac{h}{2}, w_x(t_i) + \tfrac{1}{2}k_{x_2}, w_y(t_i) + \tfrac{1}{2}k_{y_2})$$
$$k_{x_3} = hF(t_i + \tfrac{h}{2}, w_x(t_i) + \tfrac{1}{2}k_{x_2}, w_y(t_i) + \tfrac{1}{2}k_{y_2})$$

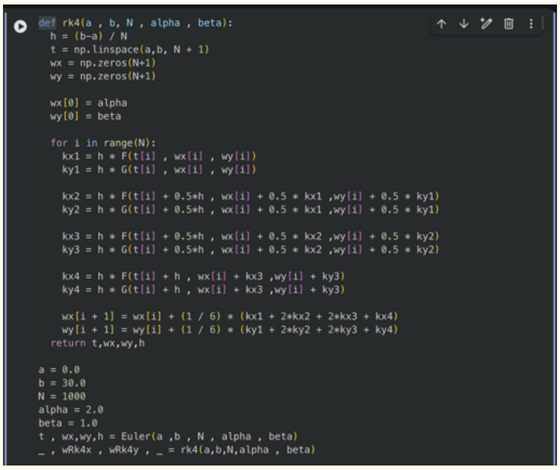
$$k_{y_4} = hG(t_i + h, w_x(t_i) + k_{x_3}, w_y(t_i) + k_{y_3})$$

$$k_{x_4} = hF(t_i + h, w_x(t_i) + k_{x_3}, w_y(t_i) + k_{y_3})$$

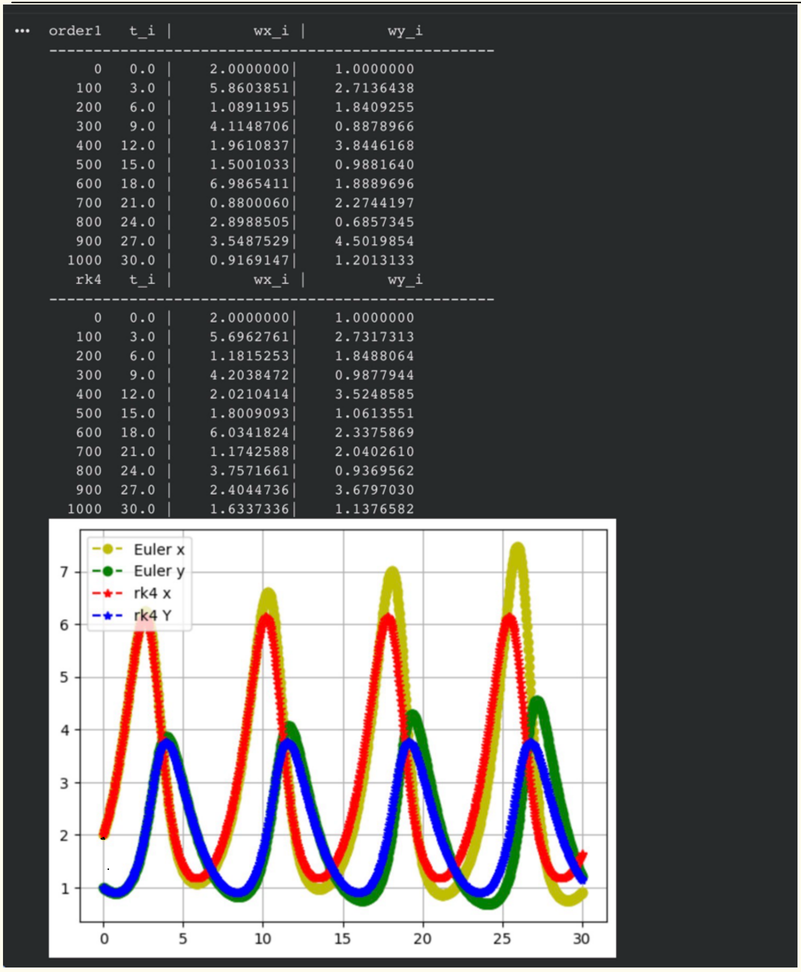
$(a,b) = (0,30)$, $N = 1000$.



```
   rk4   t_i |          wx_i |          wy_i
---------------------------------------------
     0    0.0 |     2.0000000|     1.0000000
   100    3.0 |     5.6962761|     2.7317313
   200    6.0 |     1.1815253|     1.8488064
   300    9.0 |     4.2038472|     0.9877944
   400   12.0 |     2.0210414|     3.5248585
   500   15.0 |     1.8009093|     1.0613551
   600   18.0 |     6.0341824|     2.3375869
   700   21.0 |     1.1742588|     2.0402610
   800   24.0 |     3.7571661|     0.9369562
   900   27.0 |     2.4044736|     3.6797030
  1000   30.0 |     1.6337336|     1.1376582
```



```
def rk4(a , b, N , alpha , beta):
  h = (b-a) / N
  t = np.linspace(a,b, N + 1)
  wx = np.zeros(N+1)
  wy = np.zeros(N+1)

  wx[0] = alpha
  wy[0] = beta

  for i in range(N):
    kx1 = h * F(t[i] , wx[i] , wy[i])
    ky1 = h * G(t[i] , wx[i] , wy[i])

    kx2 = h * F(t[i] + 0.5*h , wx[i] + 0.5 * kx1 ,wy[i] + 0.5 * ky1)
    ky2 = h * G(t[i] + 0.5*h , wx[i] + 0.5 * kx1 ,wy[i] + 0.5 * ky1)

    kx3 = h * F(t[i] + 0.5*h , wx[i] + 0.5 * kx2 ,wy[i] + 0.5 * ky2)
    ky3 = h * G(t[i] + 0.5*h , wx[i] + 0.5 * kx2 ,wy[i] + 0.5 * ky2)

    kx4 = h * F(t[i] + h , wx[i] + kx3 ,wy[i] + ky3)
    ky4 = h * G(t[i] + h , wx[i] + kx3 ,wy[i] + ky3)

    wx[i + 1] = wx[i] + (1 / 6) * (kx1 + 2*kx2 + 2*kx3 + kx4)
    wy[i + 1] = wy[i] + (1 / 6) * (ky1 + 2*ky2 + 2*ky3 + ky4)
  return t,wx,wy,h

a = 0.0
b = 30.0
N = 1000
alpha = 2.0
beta = 1.0
t , wx,wy,h = Euler(a ,b , N , alpha , beta)
_ , wRk4x , wRk4y , _ = rk4(a,b,N,alpha , beta)
```



```
order1   t_i |          wx_i |          wy_i
---------------------------------------------
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   400   12.0 |     1.9610837|     3.8446168
   500   15.0 |     1.5001033|     0.9881640
   600   18.0 |     6.9865411|     1.8889696
   700   21.0 |     0.8800060|     2.2744197
   800   24.0 |     2.8988505|     0.6857345
   900   27.0 |     3.5487529|     4.5019854
  1000   30.0 |     0.9169147|     1.2013133
   rk4   t_i |          wx_i |          wy_i
---------------------------------------------
     0    0.0 |     2.0000000|     1.0000000
   100    3.0 |     5.6962761|     2.7317313
   200    6.0 |     1.1815253|     1.8488064
   300    9.0 |     4.2038472|     0.9877944
   400   12.0 |     2.0210414|     3.5248585
   500   15.0 |     1.8009093|     1.0613551
   600   18.0 |     6.0341824|     2.3375869
   700   21.0 |     1.1742588|     2.0402610
   800   24.0 |     3.7571661|     0.9369562
   900   27.0 |     2.4044736|     3.6797030
  1000   30.0 |     1.6337336|     1.1376582
```
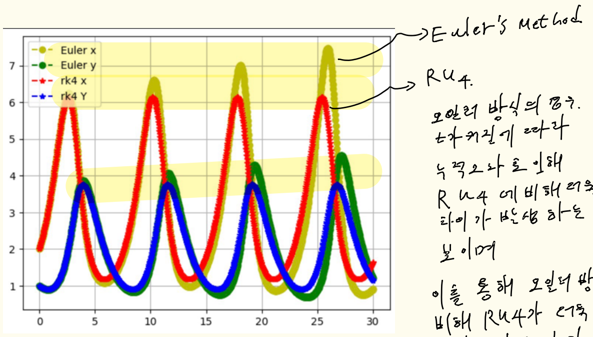

<= 오일러 방식과 RK4 비교.



오일러 방식의 경우,
t가 커짐에 따라
누적오차로 인해
RK4에 비해 더욱
차이가 발생하는 모습을
보이며

이를 통해 오일러 방식에
비해 RK4가 더욱 높은 정확도를
보임을 알 수 있다.